Can Prompt Engineering Bridge the Long-Context NLI Reasoning Gap?

Adi Pistiner, Amit Oren, Jonathan Schwartz, Yuval Zohar
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Abstract

Long-premise NLI requires locating and combining evidence across hundreds of words. We test whether question-guided decomposition can help large language models (LLMs) do this without fine-tuning. Across five models on the ConTRoL benchmark, we compare zero-shot and few-shot baselines, a retrieve-then-classify control, and six question-based configurations across three pipelines. We find no consistent advantage for question-guided decomposition over direct prompting or retrieval, and a significant disadvantage for premise-only decomposition.

1 Introduction

Natural language inference (NLI) asks whether a hypothesis is entailed by, contradicted by, or neither supported nor contradicted by a premise. Influential benchmarks such as SNLI and MultiNLI are predominantly sentence-level, and high accuracy on them does not necessarily indicate robust reasoning: models can exploit annotation artifacts and shallow syntactic heuristics that correlate with the labels (Gururangan et al., 2018; McCoy et al., 2019). Evaluating NLI over longer passages provides a stricter test, requiring models to locate and integrate evidence distributed across multiple sentences.

ConTRoL evaluates NLI over long, expert-designed passages requiring reasoning across multiple sentences and paragraphs: 8,325 premise-hypothesis pairs, premises averaging 452 words, covering logical, information-integration, coreferential, temporal, and analytical reasoning. The original paper's strongest fine-tuned baseline (BART-NLI-FT) reached only 60.95% accuracy against a 94.4% human ceiling (Liu et al., 2021), showing ConTRoL demands more than processing long input: models must identify, combine, and interpret evidence relevant to the hypothesis.

Modern LLMs can adapt to new tasks at inference time via instructions, demonstrations, or intermediate evidence-seeking steps, without ConTRoL-specific fine-tuning. Demonstrations alone can guide model behavior (Brown et al., 2020), and question-based methods can support factual verification by decomposing claims and gathering evidence (Wang et al., 2020; Min et al., 2019)—but their relative contribution to long-passage, three-way NLI is unclear. We investigate whether question-guided evidence extraction improves performance beyond direct prompting and premise retrieval alone.

We evaluate five LLMs on ConTRoL using nine method configurations. Two single-call baselines (zero-shot, few-shot) and a retrieve-then-classify control isolate the contribution of demonstrations and evidence localization. We test nine configurations. Two baselines (zero-shot, few-shot) and a retrieve-then-classify control measure how much demonstrations and evidence selection help. Six question-guided pipelines each follow the same three steps — generate questions, find and extract answers from the premise, classify the hypothesis as Entailment, Contradiction, or Neutral. The pipelines differ only in what text is used to generate the questions: the hypothesis, the premise, or both. This lets us compare which side of the premise-hypothesis pair is more useful to query.

Figure **??** shows a worked example on a real ConTRoL instance comparing three pipelines.

2 Related Work

Popular sentence-level NLI benchmarks such as SNLI and MultiNLI (Bowman et al., 2015; Williams et al., 2018) have a known problem: models can score well by learning shortcuts and surface patterns instead of actually reasoning about the premise and hypothesis (Gururangan et al., 2018; McCoy et al., 2019). Most fixes happen during

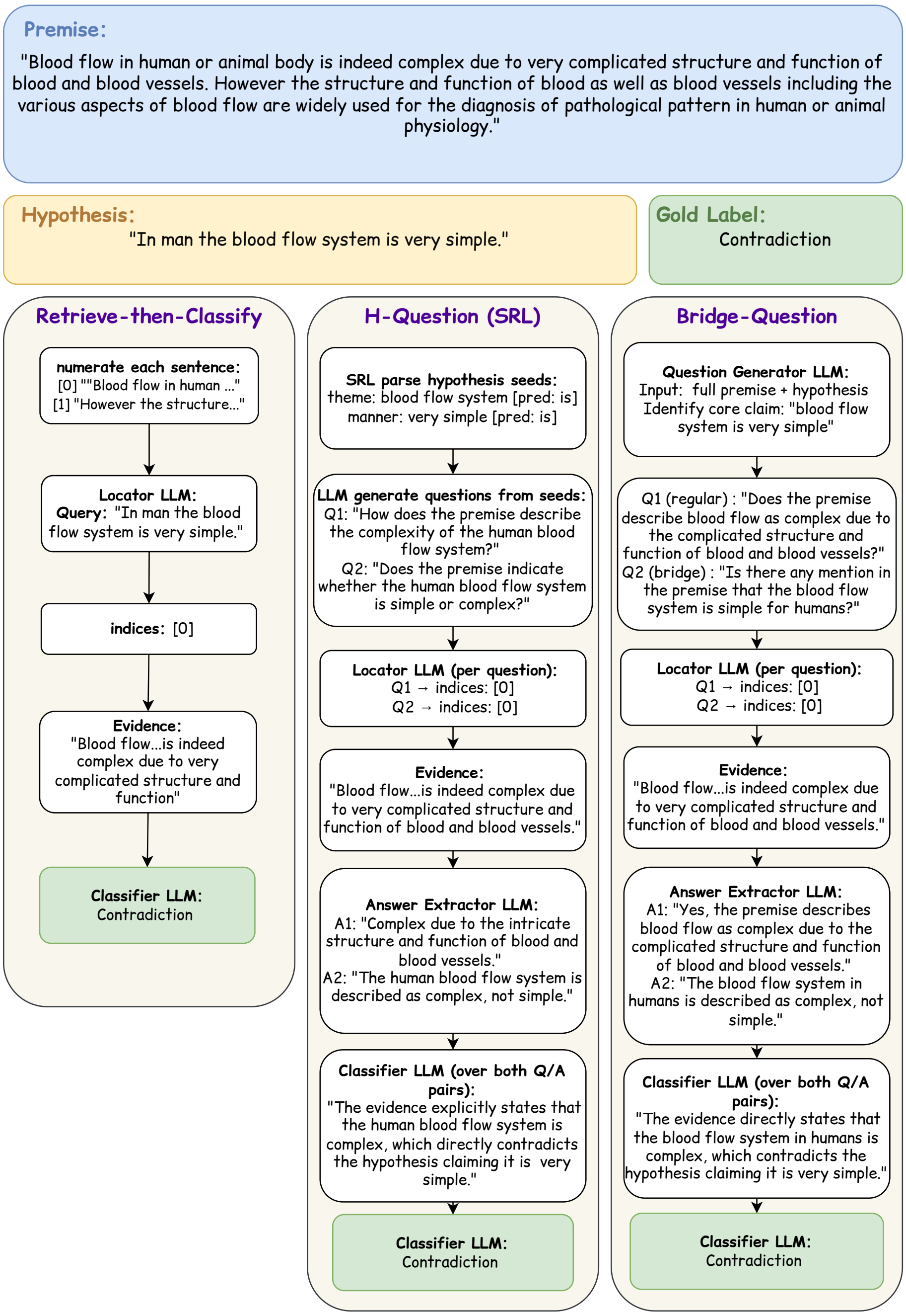

Figure 1: A worked example on a real ConTRoL instance, comparing three pipelines: RETRIEVE-THEN-CLASSIFY, H-QUESTION (SRL), and BRIDGE-QUESTION. All three correctly predict Contradiction via different evidence-extraction strategies.

training, e.g. filtering out easy examples or adding extra supervision. ConTRoL instead makes these shortcuts harder to use by requiring evidence to be found and combined across a long, multi-sentence premise (Liu et al., 2021). We take a third approach: testing whether prompting and structured question generation, at inference time with no fine-tuning at all, can close this performance gap.

In zero-shot prompting, the task is described using instructions only.few-shot in-context learning adds example demonstrations that show the model what the labels mean (Brown et al., 2020), and chain-of-thought prompting goes further by adding worked-out reasoning steps to those examples (Wei et al., 2022). Our zero-shot and three-shot conditions test only the effect of adding demonstrations: since our three-shot examples include no reasoning steps, we treat them as plain in-context learning, not chain-of-thought.

Beyond demonstrations, our question-based methods build on a different line of work: using generated questions to check and structure evidence, instead of just showing the model labeled examples. QA and NLI are closely linked tasks: Demszky et al. (2018) show that QA datasets can be mechanically turned into NLI examples, by rewriting each question-answer pair as a declarative hypothesis paired with the source passage as premise. Our pipelines lean on the reverse direction of that connection using generated questions and their answers as intermediate evidence for an NLI classifier, rather than converting them into NLI training data directly. QAGS and FEQA generate questions from a summary and answer them using the source document, to check whether it stays factually consistent (Wang et al., 2020; Durmus et al., 2020). FActScore instead breaks long generated text into small, independently checkable facts (Min et al., 2023). Multi-hop QA work such as HotpotQA and DecompRC shows the value of a related idea: finding supporting evidence and breaking a complex question into simpler ones (Yang et al., 2018; Min et al., 2019). None target three-way inference over long passages - they check summaries or answer single questions instead. We adapt their core idea to passage-level NLI: generating questions from the hypothesis, the premise, or both, and adding a retrieval-only control to show how much of the benefit comes from finding the right sentences versus the generated questions themselves.

3 Method

3.1 Dataset

We evaluate on **ConTRoL** (Liu et al., 2021), a dataset for document-level natural language inference built from expert-designed reading-comprehension material. Each instance pairs a long premise (averaging approximately 500 words) with a short hypothesis. The task is to determine whether the hypothesis is **Entailed**, **Contradicted**, or **Neutral** with respect to the premise. The dataset contains 8,325 premise-hypothesis pairs. In many cases, the evidence needed to assign a label is distributed across multiple sentences rather than concentrated in one. This makes ConTRoL more challenging than sentence-level NLI benchmarks.

3.2 Sampling

All experiments were conducted on fixed samples drawn from the ConTRoL training split using a fixed random seed (seed $= 42$).[1] The sampled instance IDs were saved once and reused identically across all models and methods, ensuring that every configuration is evaluated on exactly the same data.

Due to the higher computational cost of the premise-decomposition methods (Section 3.4), we use two sample sizes: $N = 1{,}500$ for the six methods that issue a bounded number of model calls per instance, and $N = 150$ for the three premise-question configurations, whose per-instance cost scales with the number of decomposed probe questions.

3.3 Seeding Strategies

Two of our pipelines (Section 3.4) generate probe questions from an anchor set of extracted spans, rather than directly from the raw text. We compare two strategies for constructing this anchor set.

Part-of-Speech (POS) seeding. We use NLTK's POS tagger (Loper and Bird, 2002) to extract noun phrases, main verbs, and numeric spans from the source text via shallow chunking. This method requires no model inference and is fully deterministic.

Semantic Role Labeling (SRL) seeding. Following the QA-SRL paradigm (FitzGerald et al., 2018; Pyatkin et al., 2021), we prompt the evaluated LLM to extract PropBank-style predicate-argument structures - agent (ARG0), theme (ARG1), recipient (ARG2), and modifier roles for time, location, manner, and cause - for each predicate in the source text, converting each role-argument pair into an anchor of the form `{role}: {span} [predicate: {verb}]`. This requires one extra model call, making it more expensive than POS seeding, but yields anchors with explicit predicate-argument structure instead of flat lexical spans.

[1] A naïve lexicographic sort over string-formatted instance IDs (e.g. `id_1`, `id_10`, `id_100`, ..., `id_11`) does not correspond to a numeric or random ordering and therefore yields a non-representative sample.

3.4 Methods

We evaluate nine method configurations organized into three groups. The two direct baselines are single-call: zero-shot passes the premise and hypothesis to the model directly, while few-shot uses the same single-call structure but prepends three fixed demonstrations, one worked example per label, held constant across all evaluations following Wei et al. (2022). We expect zero-shot to serve as a lower bound, as the model receives no guidance on where to look in the premise. Few-shot demonstrations should help calibrate the label format and reasoning style,but not locate evidence more precisely.

Retrieve-then-classify serves as a control that isolates the contribution of evidence localization from that of question generation. A locator model identifies the premise sentences most relevant to the hypothesis, using the **hypothesis itself** as the retrieval query; the classifier is then given these sentences verbatim, without an intermediate answer-extraction step. This condition allows any additional gain observed in the question-based methods to be attributed specifically to the generated questions, rather than to evidence retrieval alone. We expect this to outperform zero-shot by reducing the 500-word premise to a small relevant subset, but to fall short of question-based methods that structure the evidence further.

The remaining six configurations share a common three-stage architecture: (i) question generation, (ii) evidence localization and answer extraction, and (iii) classification given the resulting (question, answer) pairs. Stages (ii) and (iii) are identical across all six (detailed below), so any accuracy difference comes from which questions were generated and what text they came from, not how they are answered or judged. The three question-based methods - H-QUESTION, BRIDGE-QUESTION, and P-QUESTION - differ only in that source: the hypothesis, the premise, or both, respectively, letting us compare which side of the premise-hypothesis pair is more useful to question.

3.4.1 Answering and Classification

Locate and answer. A locator model reads the premise and one question, then picks up to five sentences from the premise that seem relevant to it. These sentences don't have to be next to each other, this lets the model combine evidence scattered across different parts of a long premise (multi-hop reasoning) instead of being limited to one contiguous block of text. An extractor model then tries to answer the question using only those five sentences. If no answer is found, the extractor drops the question instead of guessing.

Classify. The surviving question-answer pairs and the hypothesis are given to the classifier in one call, which returns a single label: Entailment, Contradiction, or Neutral. When evidence conflicts, contradiction wins over entailment, since one genuine contradiction rules out the hypothesis regardless of other supporting evidence. Neutral is used only when the evidence truly says nothing about the hypothesis.

3.4.2 Question Generation

- **h-question** generates 2-4 probe questions directly from the hypothesis, seeded via POS or SRL extraction (Section 3.3) applied to the hypothesis text. This pipeline is hypothesis-aware but premise-blind at generation time. Because generation is hypothesis-only and premise-blind, we expect it to ask relevant questions but sometimes miss the answer when the premise uses different wording.

- **bridge-question** generates 2–4 probe questions from the premise and hypothesis jointly. At least one question is required to explicitly test the hypothesis's core claim against evidence in the premise, following the bridging and comparison framing of multi-hop QA (Yang et al., 2018; Min et al., 2019). This is the only configuration whose question-generation stage observes both texts simultaneously. Because it observes both texts jointly, we expect it to generate the most targeted questions and to be the strongest of the question-based methods.

- **p-question (decomposition)** applies *atomic decomposition* to the premise: the model breaks it into the smallest standalone factual claims it contains – each covering exactly one date, name, number, or event following the atomic-fact paradigm of FActScore (Min et al., 2023). For example, "the government paid $1 lakh to families of railway-accident victims" yields separate facts for the amount, the recipients, and the triggering event. A second step identifies *relations* between facts- comparisons, causal links, and temporal orderings—following Goyal and

Durrett (2020, 2021), and each question is tagged as targeting a fact or a relation. This configuration is premise-aware but hypothesis-blind at generation time. Because it ignores the hypothesis at generation time, we expect it to produce many questions that are factually grounded but irrelevant to the specific claim being tested.

- **p-question (seeded)** generates one question per anchor extracted from the premise via the seeding strategies described in Section 3.3, evaluated separately for POS and SRL seeding. We expect SRL seeding to produce more structured, predicate-grounded questions than POS seeding, but at the cost of one extra model call per sample.

For all six question-based configurations, we additionally added three worked examples in context, following the same few-shot design. For p-question, the number of generated questions is capped at 20 per sample.

3.5 Models

We evaluate five LLMs on ConTRoL: **Qwen2.5-32B** and **gpt-oss-20B** (self-hosted, Ollama-backed), **GPT-4o-mini** and **DeepSeek-V4-Flash** (provider APIs), and **Gemma-3-27B** (via OpenRouter). This spans model scale and serving infrastructure while keeping the pipeline, prompts, and sampled data identical across all five.

4 Experimental Evaluation

4.1 Experimental Setting

We ask whether adding a question-generation step improves three-way NLI accuracy on ConTRoL beyond what zero-shot prompting, few-shot prompting, or retrieval-only classification already achieve. We report accuracy as the primary metric, together with macro-averaged recall and F1, which handle unequal label counts better than accuracy alone.

Table 1 reports accuracy across the six $N = 1{,}500$ methods and five models, with each model's best-accuracy method marked in bold. Table 2 reports macro-recall and macro-F1 for the same grid - the bolded method also scores highest under both metrics, so the ranking holds under recall and F1 too, not just accuracy. Two patterns stand out.

First, no single method wins across all models. DeepSeek-V4 baselines (zero-shot 74.6%, few-shot 73.5%) beat every question-based method on that model, including bridge-question (71.9%). The same pattern shows up across all five models: question-guided evidence extraction does not reliably beat direct prompting or retrieval-only classification here.

Second, bridge-question is consistently the strongest of the question-based methods - the only one that sees the premise and hypothesis together.

A likely reason for this narrow performance range is the pipeline design, not chance. Every question-based method passes through the same Locator/Extractor, which narrows the 500-word premise to a small number of sentences regardless of which question is asked. On Qwen2.5-32B, h-question keeps only 1.4-1.5 answered questions per sample, and bridge-question keeps 2.0; even p-question's larger question sets (13-14 per sample after filtering) go through the same narrowing step. This is close to what retrieve-then-classify already does directly - select a small relevant part of the premise - so generating a question first often adds little over retrieving on the hypothesis alone.

Seeding helps within this picture by restricting what gets asked: POS seeding ties questions to concrete noun phrases, verbs, and numbers; SRL seeding ties them to labeled predicate-argument roles. However, seeding cannot fix a Locator call that selects the wrong sentences, regardless of how well-targeted the question was.

4.2 Premise-Question Methods: An Independent Comparison

As p-question was evaluated at $N = 150$ rather than $N = 1{,}500$ (Section 3.2), Table 3 restricts every method to this identical 150-item subset for a fair, matched comparison. The performance gap widens rather than narrows here: each model's strongest baseline scores higher on this subset than on its full 1,500-sample average (e.g., DeepSeek-V4-Flash's zero-shot accuracy rises from 74.6% to 80.0%). A McNemar exact test (Dror et al., 2018) confirms this gap is not attributable to sampling noise: all three best-baseline-versus-p-question differences are significant at $p < 0.01$.

The bottleneck is question generation, not answering: generated questions are answered correctly 96–100% of the time, but 21–79% of samples never yield a usable question from Stage 1, triggering a fallback. A manual check agrees, 3 of 4 examined failures stemmed from missing questions, not wrong answers.

Model	zero-shot	few-shot	retrieve-classify	h-quest. (pos)	h-quest. (srl)	bridge-question
Qwen2.5-32B	67.1	**69.6**	69.4	67.1	66.0	69.0
gpt-oss-20B	**70.8**	70.0	68.2	64.7	65.9	67.7
GPT-4o-mini	65.9	65.5	**67.1**	59.8	–	62.5
DeepSeek-V4-Flash	**74.6**	73.5	71.3	59.4	69.3	71.9
Gemma-3-27B	64.5	64.9	63.5	64.5	63.2	**67.0**

Table 1: Accuracy (%) across the six $N = 1{,}500$ methods and five models. '–' marks GPT-4o-mini's pending h-question (srl) rerun (Section 4.1).

Model	zero-shot	few-shot	retrieve-classify	h-quest. (pos)	h-quest. (srl)	bridge-question
Qwen2.5-32B	68.0/67.2	**69.8/69.5**	69.5/69.2	67.1/66.9	65.8/65.7	68.5/68.5
gpt-oss-20B	**71.4/70.6**	70.5/69.9	68.7/67.7	64.3/64.4	65.6/65.5	67.1/67.1
GPT-4o-mini	65.3/65.3	65.2/65.1	**66.3/66.1**	57.3/55.3	-	59.9/58.1
DeepSeek-V4-Flash	**75.2/74.6**	74.0/73.4	71.9/71.2	60.2/59.8	68.7/68.6	71.0/71.0
Gemma-3-27B	64.2/63.4	64.0/63.5	62.5/61.8	63.2/62.9	61.9/61.5	**65.7/65.3**

Table 2: Macro-Recall / Macro-F1 (%) for the same six methods and five models as Table 1, shown as R/F1 in each cell.

4.3 Ablation and Error Analysis

Question count. Our main sweep fixes h-question at two to four probe questions per sample. A separate ablation raised this to six to eight, holding everything else constant. The effect was strongly model-dependent: Qwen2.5-32B barely moved (67.1% to 67.5%, +0.4pp), whereas DeepSeek-V4-Flash gained substantially (59.4% to 66.8%, +7.4pp). The same change thus helped one model and left another essentially unaffected.

A concrete generation failure. For the premise "All boxes are cartons. All cartons are packages. Some packages are letters. No box is a parcel" and hypothesis "Some boxes are not cartons" (gold: Contradiction), p-question's decomposition step (Stage 1) returned zero questions. This premise is a short categorical syllogism with no long-range dependencies, so the failure cannot be attributed to premise length or complexity: the generation prompt itself, rather than input difficulty, appears to drive it. The pipeline fell back to zero-shot, which happened to predict correctly here – masking the Stage 1 failure at the level of final accuracy, and illustrating why our fallback-rate diagnostic is necessary.

Aggregation and fallback. Before adopting the classifier of Section 3.4, we tried a simpler aggregation scheme: each answered question casts one vote for a label, and the majority wins. It failed on a clear case. A sample with ten answered questions contained one decisive answer – a direct contradiction that settles the label on its own – but it was outvoted 9-to-1 by answers that were individually correct and genuinely Neutral, yet irrelevant to the claim under test. Majority voting counts evidence without weighing it, and so cannot separate "correct and decisive" from "correct but beside the point." Our classifier instead treats one sufficiently decisive answer as enough to determine the label, regardless of how much unrelated evidence surrounds it. When a pipeline yields no usable evidence at all, it falls back to a plain zero-shot call, and only to a hardcoded Neutral label if that also fails.

Limitations

Sample-size asymmetry. The three P-QUESTION variants are evaluated at $N = 150$ rather than $N = 1{,}500$ (Section 3.2) due to higher per-sample cost.

Limited significance testing. Each method and model was run once, so most reported numbers had no confidence interval. Only the P-QUESTION comparison was tested for significance (McNemar).

Overlap-driven fallback is a hypothesis, not a confirmed mechanism. ConTRoL has low hypothesis-premise lexical overlap (4.87–5.49%; Liu et al., 2021), and this overlap correlates with H-QUESTION's fallback rate (Mann-Whitney U, $p < 10^{-4}$ across all four models tested). This is consistent with the Extractor sometimes missing paraphrased answers, rather than the evidence being truly absent – but we have not confirmed it

Model	zero-shot	few-shot	retrieve-classify	h-quest. (pos)	bridge-question	p-quest. (decomp)	p-quest. (seed,pos)	p-quest. (seed,srl)
Qwen2.5-32B	70.0	74.7	78.0†	74.7	73.3	59.3	61.3†	60.7
DeepSeek-V4-Flash	80.0†	80.0	79.3	71.3	77.3	68.7†	62.7	62.0
Gemma-3-27B	68.0	68.7	64.0	68.0	74.0†	55.3†	47.3	51.3

Table 3: Accuracy (%) on the same 150-item subset used by the p-question methods, so that every method is compared fairly against p-question despite its smaller N (Section 3.2). † marks the per-model best-baseline/best-p-question pair tested with a McNemar exact test (Dror et al., 2018); all three pairs are significant at $p < 0.01$.

by manually inspecting cases, so we report it as a hypothesis rather than a finding.

No human evaluation of intermediate outputs. Our fallback and answerable-rate diagnostics measure pipeline behavior automatically, some failure modes beyond "no question generated" or "unanswerable" may go undetected without human review.

5 Conclusion

We ask whether question-guided prompting can improve three-way NLI (Entailment, Contradiction, Neutral) on the long-context ConTRoL benchmark, using only inference-time methods with no fine-tuning. Across five instruction-tuned models, question-based pipelines do not consistently beat zero-shot, few-shot, or retrieve-then-classify baselines: a baseline or retrieval control wins on four of the five models. The six methods score close enough that we treat their ranking as a hint, not a firm result.

A shared retrieval step explains the null result. One possible explanation for why the question-based pipelines do not beat the baselines is the design they share, though we cannot rule out other factors: every method sends its output through the same locator, which reduces the premise (averaging 500 words) to a small number of sentences before the classifier ever sees it-close to what retrieve-then-classify already does directly –so generating a question first often adds little. This shows up in how few questions survive to be used as evidence: h-question keeps only 1.4–1.5 answered questions per sample, and bridge-question keeps about 2.0.

Premise-only decomposition fails at generation, not answering. Premise-only decomposition (p-question) is the one method that performs significantly worse, confirmed by a real statistical test (McNemar, $p < 0.01$). The problem isn't answer quality, looking only at the individual questions that do get generated, 96–98% of them are answered correctly (Qwen2.5-32B). The problem is earlier: many samples never generate a single usable question to begin with, and 21–39% end up with nothing at all, forcing a fallback- a plain zero-shot call as backup, or a hardcoded Neutral guess if even that fails. A related pattern shows up in h-question: when it fails to answer a question, it's usually because the hypothesis and premise barely share any of the same words, suggesting the model sometimes misses an answer that's just phrased differently rather than truly absent.

Joint context visibility and verbatim evidence help most. Among the question-based methods, bridge-question, which sees both premise and hypothesis, works best every time, and retrieving whole premise sentences beats breaking the premise into separate questions across every model. Both results point to the same idea: keeping evidence intact preserves the logical connections between sentences, while splitting it into isolated question-answer pieces can break them apart.

Modern zero-shot narrows the original gap. The strongest models already reach accuracies well above this dataset's original fine-tuned baseline (BART-NLI-FT: 60.95%; Liu et al., 2021) using zero-shot prompting alone. This isn't conclusive evidence against question generation, though: most gaps between methods haven't been tested for significance, and one of our own experiments raised DeepSeek-V4 accuracy by 7.4% just by asking more questions.

Outlook. It's still an open question whether asking more, or better-targeted, questions would change these results: our nine methods vary in question source (hypothesis, premise, or both), not question count, so the two effects can't be separated here. Future work should explore adaptive mechanisms that ask more questions when evidence is inconclusive, rather than relying on a fixed budget, and retrieval methods that preserve logical connections.

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